

GAUTAM SOLAR

Solar Module Product Datasheet 2024

1. N-Type TOPCon Solar Modules

Gautam Solar's N-Type TOPCon modules represent the pinnacle of photovoltaic technology. Using tunnel oxide passivated contact architecture, these modules deliver exceptional efficiency and superior performance in real-world conditions including low-light and high-temperature environments.

Parameter	G12 R TOPCon (640Wp)	N-Type TOPCon (600Wp)
Peak Power (Wp)	620–640	590–600
Module Efficiency (%)	22.95–23.69	Up to 22.84
Open Circuit Voltage (Voc)	38.6 V	37.8 V
Short Circuit Current (Isc)	18.2 A	17.5 A
Max Power Voltage (Vmp)	32.4 V	31.9 V
Max Power Current (Imp)	17.6 A	16.8 A
Bifacial Gain	Up to 30%	Up to 25%
Temperature Coefficient (Pmax)	-0.29%/°C	-0.30%/°C
Operating Temperature	-40°C to +85°C	-40°C to +85°C
Dimensions (mm)	2278×1134×30	2172×1303×30
Weight (kg)	32.5	28.0
Frame	Anodized Aluminium	Anodized Aluminium
Connector Type	MC4 Compatible	MC4 Compatible
Warranty — Product	12 Years	12 Years
Warranty — Performance	30 Years Linear	25 Years Linear

2. Mono PERC Solar Modules

Our Mono PERC modules combine M10 Half-Cut Cell technology with MBB (Multi-BusBar) design and Round Ribbon Connectors. Manufactured using NDC (Non-Destructive Cutting), these modules offer excellent performance for residential, commercial, and utility-scale applications.

Parameter	Bifacial G-B (550Wp)	MonoFacial (550Wp)
Peak Power (Wp)	540–550	540–550
Module Efficiency (%)	20.9–21.3	20.5–21.0
Open Circuit Voltage (Voc)	49.8 V	49.5 V
Short Circuit Current (Isc)	13.9 A	13.7 A
Bifacial Gain	Up to 20%	N/A (MonoFacial)
Temperature Coefficient (Pmax)	-0.35%/°C	-0.35%/°C

Dimensions (mm)	2278×1134×30	2278×1134×35
Warranty — Product	12 Years	12 Years
Warranty — Performance	25 Years Linear	25 Years Linear

3. Certifications & Compliance

- BIS (Bureau of Indian Standards) — IS 14286 Certified
- IEC 61215 & IEC 61730 — International Safety & Performance Standards
- UL 61730 — Underwriters Laboratories Certification (USA Market)
- TUV Rheinland — Product Safety Certification
- CE Marking — European Conformity
- SGS Certified — Independent Quality Verification
- MNRE ALMM Listed — Approved List of Models and Manufacturers (India)
- PID Free — Potential Induced Degradation Resistant
- Salt Mist & Ammonia Resistant — IEC 61701 & IEC 62716

4. Performance Degradation Warranty

Year	Min. Power Output (%)	Annual Degradation
Year 1	98.0%	2.0% (initial)
Year 5	96.4%	0.40%/year
Year 10	94.4%	0.40%/year
Year 15	92.4%	0.40%/year
Year 20	90.4%	0.40%/year
Year 25	88.4%	0.40%/year
Year 30 (TOPCon)	86.4%	0.40%/year